# SAAHIL KHAN MEHAR

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## PROFESSIONAL SUMMARY

B.Tech Computer Science & Engineering student (graduating 2027) with hands-on experience in Python, FastAPI and PostgreSQL. Practical background in REST API design, JWT/OAuth 2.0 authentication, Redis, Docker and AWS. Experienced in building tested, well-documented backend services (Pytest, Swagger/OpenAPI, CI/CD) and in leading a final-year project team.

## TECHNICAL SKILLS

**Programming Languages:** Python, SQL, JavaScript, HTML, CSS
**Backend Frameworks:** FastAPI, REST API Development, Pydantic, Uvicorn
**Databases & ORM:** PostgreSQL, MySQL, Redis, SQLAlchemy, Alembic
**API & Security:** JWT Authentication, OAuth 2.0, CRUD Operations, API Integration, Swagger/OpenAPI
**Testing:** Pytest, Unit Testing, Integration Testing, Postman
**Tools & Platforms:** Git, GitHub, Docker, Linux, VS Code
**Cloud & DevOps:** AWS (EC2, S3, IAM), CI/CD Pipelines
**AI Coding Tools:** OpenCode, Claude Code, Google Antigravity

## BACKEND & TECHNICAL PROJECTS

**CareerOPS: Job-search backend API that tracks applications and scores job descriptions**
*Python 3.12, FastAPI, SQLAlchemy 2.0 (async), PostgreSQL, Redis, Celery, Alembic, JWT, Pytest*

- **Problem:** Job seekers lose track of applications and can't see which skills the market actually asks for.
- **What I Built:** Layered FastAPI service with an application state machine, deterministic 0–100 JD match scoring, a Celery pipeline ingesting Greenhouse, Lever and RemoteOK listings, SQL analytics cached in Redis, and JWT + API-key auth with rate limiting.
- **Proof It Worked:** 166 automated tests with 98.4% coverage; skill extraction scored 1.0 precision/recall on 20 labeled JDs; used daily to run my own job search.

**PayLedger API: Double-entry payment ledger with idempotent transfers**
*Python, FastAPI, PostgreSQL, JWT (RBAC), APScheduler, Pytest*

- Built a double-entry ledger API with balanced debit/credit entries enforced by a database constraint, and idempotent transfers so client retries never double-post, even under concurrent requests.
- Added admin/accountant/viewer RBAC, an append-only audit log, and a reconciliation worker that detects ledger discrepancies.

**PulseBoard API: Real-time product-analytics event ingestion**
*Python, FastAPI, PostgreSQL, Redis Streams, WebSockets, Pytest*

- Built a batch event-ingestion API with API-key auth and rate limiting; Redis Streams consumer groups feed an async worker that bulk-writes to PostgreSQL, with a direct-write fallback if Redis is down.
- Added minute/hourly rollups with retention pruning and a JWT-authenticated WebSocket live feed; 27 tests, Locust load test and CI.

## EDUCATION & ACADEMIC CREDENTIALS

**B.Tech, Computer Science & Engineering** *CGPA: 8.10 / 10.0*
CMR University, Bengaluru Graduating Batch: 2027

## PROFESSIONAL EXPERIENCE

**Intern | State Bank of India (SBI), Vidya Nagar Branch** *4 months*

- Handled day-to-day branch banking operations, including internet banking activation and KYC updates, with accuracy and attention to customer data.
- Received an "Excellent" conduct rating and a completion certificate from the branch.

## HONORS & CERTIFICATIONS

- **Hackathons** — 10+ hackathons: built and presented working prototypes in team settings under tight deadlines.
- **Machine Learning** (NPTEL, IIT Kharagpur): completed the NPTEL certification course in Machine Learning.
- **Final Year Project Lead** — Vivada Setu, CMR University: leading the team building Vivada Setu, a consumer court project, as our B.Tech final-year project.